

Math 63: Real Analysis

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Problem 1. Let $S \subset E$ be a subset of a metric space E , and let $C \subset E$ be a closed subset of E that contains S . Prove that C contains all cluster points of S .

Solution.

Note that,

$$S \subset C \subset E$$

By the Theorem on Page 47, we know that a subset $C \subset E$ is closed if and only if C contains the limits to all sequences of C (and hence, all limits of sequences of S) that are convergent in E . Recall from HW4, Problem 1 that $\mathbf{p}$ is a cluster point of S if and only if $\mathbf{p}$ is the limit of a Cauchy sequence of points in $S \cap \{\mathbf{p}\}^c$. That is, all cluster points of S are limits to Cauchy sequences in S . Then, because C is closed it must contain all cluster points of S .

Problem 2. Two sequences $p_1, p_2, p_3, \dots$ and $q_1, q_2, q_3, \dots$ are called equivalent if for every $\varepsilon > 0$ there exists a positive integer N such that

$$d(p_n, q_n) < \varepsilon \quad \text{for all integers } n > N.$$

Prove that if (p_n) and (q_n) are equivalent sequences, then (p_n) converges if and only if (q_n) converges.

Solution.

($\implies$)

Assume (p_n) and (q_n) are equivalent sequences, and that (p_n) converges.

Let p be the limit point of (p_n) . Then, note that for every $\varepsilon/2 > 0$ there exists positive integers N_1 , and N_2 such that for $N = \max\{N_1, N_2\}$, for all integers $n > N$, the following two inequalities hold,

$$d(p_n, q_n) < \frac{\varepsilon}{2}, \quad d(p_n, p) < \frac{\varepsilon}{2}$$

Then by the triangle inequality,

$$d(q_n, p) < d(p_n, q_n) + d(p_n, p) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

That is, for each $\varepsilon > 0$ we have a positive integer N such that for all integers $n > N$,

$$d(q_n, p) < \varepsilon$$

That is, (q_n) converges to p .

($\impliedby$)

Assume (p_n) and (q_n) are equivalent sequences, and that (q_n) converges. (p_n) converges by a symmetric argument to the one above, and we get the following result: For each $\varepsilon > 0$ we have a positive integer N such that for all integers $n > N$,

$$d(p_n, q) < \varepsilon$$

That is, (p_n) converges to q .

Hence, if (p_n) and (q_n) are equivalent sequences, then (p_n) converges if and only if (q_n) converges.

Problem 3. Let E be the metric space defined in Chapter III, Problem 1(b). Let B be the closed unit ball in E with center $(0, 0, 0, \dots)$ and radius 1. Then B is closed and bounded. Prove that B is not compact.

Hint. Construct a sequence in E that does not have a convergent subsequence.

Hint 2. You do not have to use the hints.

Solution.

The metric space defined in Chapter III, Problem 1(b) is,

$$E = \{\text{all bounded infinite sequences } x = (x_1, x_2, x_3, \dots) \text{ of elements of } \mathbb{R}\}$$

with

$$d(x, y) = \text{l.u.b.}\{|x_1 - y_1|, |x_2 - y_2|, |x_3 - y_3|, \dots\}$$

Let B be a closed unit ball in E centered at $(0, 0, 0, \dots)$.

$$B = B_1(0, 0, 0, \dots)$$

Consider the sequence (p_n) formed by points $p_i = (0, \dots, 0, 1, 0, \dots)$ where the i -th entry of the tuple is 1. Then the distance between any two such points p_i, p_j is exactly 1 since,

$$d(p_i, p_j) = \text{l.u.b.}\{|p_{i_1} - p_{j_1}|, |p_{i_2} - p_{j_2}|, \dots, |p_{i_i} - p_{j_i}|, \dots, |p_{i_j} - p_{j_j}|, \dots\}$$

and the value of the element corresponding to a term in the set is 1 for the terms $|p_{i_i} - p_{j_i}|$ and $|p_{i_j} - p_{j_j}|$ and zero otherwise, that is, $d(p_i, p_j) = 1$, for any points p_i, p_j in the sequence (p_n) . That is, every pair of point in the sequence is exactly 1 away from each other hence it is not a Cauchy sequence. Since all convergent sequences are Cauchy in metric spaces (Proposition on Page 51), this sequence is not convergent and hence B is not compact (by Corollary 1 on Page 56).

Problem 4. A curve γ in a metric space E is a continuous function

$$\gamma : [0, 1] \rightarrow E.$$

Suppose that for any two points $\mathbf{p}, \mathbf{q} \in E$ there exists a curve γ with $\gamma(0) = \mathbf{p}$ and $\gamma(1) = \mathbf{q}$. Prove that E is connected.

Solution.

Fix a point $\mathbf{p}$ in E . Then for any point $\mathbf{q}$ in E define $\gamma_{\mathbf{q}} : [0, 1] \rightarrow E$ such that $\gamma_{\mathbf{q}}(0) = \mathbf{p}$ and $\gamma_{\mathbf{q}}(1) = \mathbf{q}$. Note that $[0, 1]$ is connected (Theorem on Page 60), and since γ is a continuous function $\gamma_{\mathbf{q}}([0, 1])$ is also connected (Theorem on Page 78). Then E can be represented as a union of connected sets

$$E = \bigcup_{\mathbf{q} \in E} \gamma_{\mathbf{q}}([0, 1])$$

with a non-empty intersection at $\mathbf{p}$. Then by the Proposition on Page 60, E is connected.

Problem 5. If γ_1 and γ_2 are two curves in E with $\gamma_1(1) = \gamma_2(0)$, then the concatenation of γ_1 and γ_2 is the curve

$$\gamma_3 : [0, 1] \rightarrow E$$

defined by

$$\gamma_3(t) = \begin{cases} \gamma_1(2t), & 0 \leq t \leq \frac{1}{2}, \\ \gamma_2(2t - 1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

Note that γ_3 is well-defined because if $t = \frac{1}{2}$ then $2t = 1$ and $2t - 1 = 0$, so $\gamma_1(2t) = \gamma_2(2t - 1)$.

Prove that γ_3 is continuous at $t = \frac{1}{2}$.

Solution.

Let $d(p, q) = |p - q|$ be the distance metric on $[0, 1]$ and $d'(p, q)$ be the distance metric on E' . Since γ_1 and γ_2 are continuous functions (curves) in E , by the $\epsilon - \delta$ definition of continuity for a given ϵ , there exist $\delta_1, \delta_2 > 0$ such that for any point $p \in [0, 1]$,

$$d(p, 1) = |1 - p| < \delta_1 \implies d'(\gamma_1(p), \gamma_1(1)) < \epsilon$$

and,

$$d(p, 0) = |0 - p| < \delta_2 \implies d'(\gamma_2(p), \gamma_2(0)) < \epsilon$$

Now define $\delta = \frac{1}{2} \min\{\delta_1, \delta_2\}$. Then, for some point $t \in [0, 1]$ if $|t - \frac{1}{2}| < \delta$, then either $t \geq \frac{1}{2}$ or $t \leq \frac{1}{2}$.

If $t \leq \frac{1}{2}$, then $|2t - 1| < \delta_1$. Also note that, $\gamma_3(t) = \gamma_1(2t)$,

$$d'(\gamma_3(t), \gamma_3(\frac{1}{2})) = d'(\gamma_1(t), \gamma_1(1)) < \epsilon$$

and if $t \geq \frac{1}{2}$, then $|2t - 1 - 0| < \delta_2$. Also note that, $\gamma_3(t) = \gamma_2(2t - 1)$,

$$d'(\gamma_3(t), \gamma_3(\frac{1}{2})) = d'(\gamma_2(t), \gamma_2(0)) < \epsilon$$

That is, γ_3 is continuous at $t = \frac{1}{2}$.